

NETWORK INTERFACE AND SECURITY ENHANCEMENTS – [NIC]

❖ **Network Interface Card (NIC) – Elastic Network Interface (ENI)**

- In AWS, a Network Interface Card **is called an** Elastic Network Interface (ENI).
- It is a virtual network component that can be **attached** to an EC2 instance to **provide network connectivity**.

➤ **Each ENI includes:**

- A *private IPv4* address (mandatory)
- One or more *secondary private IP* addresses (optional)
- One Elastic IP address (optional)
- A MAC address
- One or more security groups

ENIs can be attached or detached from EC2 instances. This allows you to move network interfaces across instances without changing the network configuration.

✓ **ENIs are useful in scenarios like:**

- Creating instances with multiple network interfaces (e.g., firewall, router)
- Moving network interfaces to new instances during failover
- Preserving network settings across instance replacements

❖ **Elastic IP (EIP)**

- An Elastic IP is a static public IPv4 address that you can **allocate** to your AWS account.
- Unlike automatically assigned public IPs, an Elastic IP remains the same even when you **stop** or **restart** your EC2 instance.

➤ *You can associate an Elastic IP with:*

- An EC2 instance directly
- An ENI, which is then **attached** to an instance

➤ **Elastic IPs are useful when:**

- You want a fixed public IP for your server
- You need to remap your IP quickly to another instance during failover
- You need to whitelist your IP in external systems

✓ **Key Difference:**

- ENI is a virtual network card that **connects your instance** to the network.
- Elastic IP is a **static public IP** address that gives your instance or ENI **a fixed point** of contact on the **internet**.

Together, ENI and Elastic IP help build flexible, reliable, and scalable cloud networking solutions.

❖ **Placement Group?**

*A **Placement Group** is a **logical grouping of EC2 instances** within a single AWS Region that determines **how instances are placed on underlying hardware**.*

🎯 **Types of Placement Groups (Important for Interviews)**

1. **Cluster Placement Group**

*Places instances **close together** on the **same physical hardware** within a single Availability Zone.*

- **Use Case:** Low-latency, high-throughput workloads (e.g., HPC, tightly-coupled systems).
- **Pros:** Very fast communication (10 Gbps+).
- **Limits :-** Limited by AZ capacity

2. Spread Placement Group

Places each instance on **distinct hardware** (different racks, power sources).

- **Use Case:** High availability; critical instances (e.g., database nodes).
- **Pros:** Reduces risk of simultaneous failures.
- **Limits :-** Max 7 instances per AZ per group (soft limit).

3. Partition Placement Group

- **Definition:** Divides **instances into partitions**; each partition is isolated from hardware failures of others.
- **Use Case:** Large-scale distributed and replicated workloads (e.g., HDFS, Cassandra).
- **Pros:** Isolation within group; supports thousands of instances.
- **Limits :-** Must plan partitions properly.

Type	Cluster	Partition	Spread
Description	Places instances close together in a single AZ	Divides instances across partitions (logical groups), each in separate racks	Spreads instances across multiple hardware racks
Use Case	Low-latency, high-throughput apps (e.g., HPC, big data)	Large-scale distributed apps like Hadoop, Cassandra	Critical, small number of instances (e.g., DNS, web front-end)
Max instances	No limit (but constrained by AZ capacity)	Depends on partitions (up to 7 per partition per AZ)	7 per AZ
Fault tolerance	✗ Low	✓ Medium	✓ high
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Fault tolerance	✗ Low	✓ Medium	✓ high

NACL vs Security Group

Both **NACLs** and **Security Groups** are used to control network traffic in AWS, but they work at different levels and have different behaviors.

❖ Security Group

✓ A Security Group is a virtual firewall that controls inbound and outbound traffic for EC2 instances.

➤ Key Characteristics:

- **Works at the instance level** (attached to the ENI of the instance).
- **Stateful :-** Return traffic is automatically allowed, even if not explicitly defined.
- **Default behavior :-** All inbound traffic is denied, all outbound traffic is allowed (can be modified).

- **Only allows traffic** :— cannot explicitly deny.
 - **Rules** are based on protocol, port, and source/destination IP or Security Group.
 - **Use Case** :- Fine-grained control over what traffic is allowed to or from a specific instance.
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- **Network ACL (NACL)**
 - ✓ A Network Access Control List (NACL) is an optional layer of security that acts as a firewall at the subnet level.
 - ✓ **Key Characteristics:**
 - **Works** at the subnet level, affecting all resources within the **subnet**.
 - **Stateless** :- Return traffic must be explicitly allowed in the rules.
 - **Default behavior** :- All inbound and outbound traffic is allowed in the default NACL. In custom NACLs, all traffic is denied until rules are added.
 - **Allows and denies** traffic explicitly.
 - Rules are evaluated in numbered order (from lowest to highest).
 - **Use Case** :- Broad control over traffic entering or leaving a subnet, such as blocking IP ranges or setting up subnet-wide restrictions.
 - **NOTE:-**
 - **Stateless:-** Does **not store** any information about previous requests.
 - **Statefull :-** Remembers past interactions or session data.
 - **Comparission SG & NACL:**

Feature	Security Group	Network ACL (NACL)
Level of Operation	Instance-level (attached to EC2)	Subnet-level (applies to all resources in subnet)
Statefulness	Stateful – return traffic is allowed	Stateless – return traffic must be explicitly allowed
Rules Applied	Only Allow rules	Allow and Deny rules
Use Case	Controls inbound/outbound traffic for EC2	Controls inbound/outbound traffic at subnet level
Applies To	ENIs (Elastic Network Interfaces)	Entire subnet

[GENERAL INTERVIEW QUESTIONS]

Q1. What is an ENI in AWS?

- **ENI** (Elastic Network Interface) is a virtual network card attached to an EC2 instance.
- It has a MAC address, private IP, public IP (optional), and security groups.

Q2. Why would you use multiple ENIs with an EC2 instance?

- Attach to multiple subnets
- Achieve network-level isolation
- Support failover or HA designs
- Separate traffic (e.g., frontend/backend)

Q3. What is an Elastic IP and when would you use it?

- Elastic IP is a static public IP address that can be attached to an EC2. Useful for persistent access even if the instance is stopped or replaced.

Q4. Can you attach an Elastic IP to a secondary ENI?

- **Yes.** Elastic IPs can be attached to any ENI with a public IP allocation.

Q5. What are the types of placement groups in AWS?

- **Cluster** – EC2s placed close together for **low** latency
- **Spread** – EC2s spread across AZs for \
- **9** availability
- **Partition** – EC2s grouped into logical partitions for distributed workloads.

Q6. Can you move an existing EC2 instance into a placement group?

- **No,** you must stop and relaunch the instance in the placement group.

Q7. Which placement group would best suit Big Data applications?

- **Partition group** – allows grouping into isolated partitions, each on separate racks.

Q8. How are NACLs and Security Groups difference in terms of scope?

Q9. How do stateful and stateless behavior difference between SG and NACL?

Q9. What happens if both NACL and Security Group conflict on rules?

- The most restrictive rule applies. If either denies the traffic, it is denied.

Q10. How does cluster placement group improve performance?

- Instances are **placed** in the **same AZ** on physically close hardware, which improves network throughput and low latency.